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ELECTRIC, FUEL CELL, AND HYBRID VEHICLE, ENGINE AND EMISSIONS, FUELS AND LUBRICANTS, HEAT TRANSFER, FLUID AND THERMAL ENGINEERING

Driving Mode Development for Concrete-Mixer Trucks Based on Suburban Power Demand Characteristics

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Abstract
As environmental regulations tighten and carbon neutrality becomes a global priority, construction equipment is increasingly transitioning toward zero-emission power sources. To ensure the feasibility of applying eco-friendly power sources, it is essential to precisely understand the power demand characteristics under real-world operating conditions. This study developed representative driving modes for a 6-rube class mixer truck based on actual driving data from suburban driving. Using a micro-trip methodology, 8,491,392 and 3,964,032 candidate driving cycles were generated for transport and return trips, respectively. Representative modes were selected based on speed distribution similarity, with low MAE values of 1.96% and 0.83%. The resulting power demand profiles showed a peak requirement of 282 kW, and moving average analysis demonstrated the potential for managing peak loads through appropriate system sizing. These representative modes and derived power characteristics can be used to support efficient and low-emission system development for mixer trucks.

Keywords Driving mode · Mixer truck · Power characteristics · Driving pattern · Micro-trip · Suburban areas

1 Introduction

Global Construction equipment consumes significantly more energy than passenger and commercial vehicles due to its operating characteristics. Consequently, conventional construction equipment has a more substantial negative environmental impact, emitting higher levels of greenhouse gases and harmful emissions per fleet compared to vehicles (Xueyao et al., 2023; Zhang et al., 2020). Projections indicate that emissions from construction equipment will surpass those of vehicles by 2030 and account for more than 80% of mobility sector emissions in East Asia, Europe, and North America by 2050 (Xueyao et al., 2023). To mitigate these emissions, various regulations have been implemented targeting construction equipment. Notable examples include the Tier regulations enforced by the U.S. Environmental Protection Agency (EPA) and the Stage regulations implemented in Europe (Fnu et al., 2024; Shao et al., 2016). The current U.S. Tier regulation is at Tier 4 Final, with updates occurring approximately every 4 to 5 years. The upcoming Tier 5 regulation, expected to be significantly stricter, is scheduled for implementation in 2028. Similarly, Europe enforces Stage V regulations, mandating compliance for engines with power ratings between 8 kW and 560 kW sold after 2020. Additionally, strategies and goals are currently being established globally for carbon neutrality. In the United States, the goal is to become carbon neutral with five key elements: decarbonizing electricity, switching to clean fuels, reducing energy waste, reducing greenhouse gas emissions, and eliminating CO_2 . In Europe, the goal is to reduce greenhouse gas emissions by 55% by 2030 and become carbon neutral by 2050 (Béres et al., 2024). Europe is also targeting zero emissions from construction sites, which account for about 40% of energy and process-related emissions (Stokke et al., 2022).

With The global trend toward stringent emission regulations and carbon neutrality, the construction equipment

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industry is increasingly being required to transition to eco-friendly electric power sources (Trinh et al., 2022). Due to the high energy consumption of construction equipment, applying a battery-electric power system is challenging, making hydrogen fuel cell electric systems an attractive alternative. However, Hydrogen fuel cells cannot be used as standalone power sources for construction equipment, unlike conventional engines. Instead, they are integrated into hybrid systems with high-voltage batteries, which have a relatively smaller capacity compared to battery-electric systems (Diana et al., 2024). To configure an optimal hydrogen fuel cell system for a specific type of mobility, it is essential to determine the power output of the hydrogen fuel cell and the capacity of the high-voltage battery by considering the specific power demand characteristics of the vehicle. Determining the power demand of mobility requires an appropriate driving mode. For vehicles, several established driving modes, such as Federal Test Procedure (FTP-75) and Worldwide Harmonized Light Duty Test Cycle (WLTC), are approved for fuel economy and emissions certification. These modes serve as valuable references for developing hydrogen fuel cell systems for vehicles (Changizian et al., 2020). In the case of construction equipment, driving modes for engine certification, such as Non-Road Steady Cycle (NRSC) and Non-Road Transient Cycle (NRTC), exist for emissions evaluation (Di Luca et al., 2022; Ettl et al., 2016; Ullman et al., 1999). However, these modes are designed as engine-based standards that reflect general power patterns across the entire non-road sector, including construction equipment. Consequently, they do not account for the unique operating characteristics of specific types of construction equipment.

To address this gap, this study focuses on concrete mixer trucks, which exhibit unique and complex driving behaviors including drum rotation during transport, dynamic weight variation due to concrete loading, and frequent acceleration/deceleration patterns that are not adequately reflected in existing cycles. Moreover, most prior research on driving mode development for heavy-duty vehicles has focused on urban freight or logistics trucks, with limited attention to mixer trucks operating in suburban environments, where most concrete plants are located. Therefore, the motivation of this study lies in developing representative driving modes based on actual driving data from mixer trucks, enabling accurate power demand estimation and providing a practical foundation for the design of hydrogen fuel cell hybrid systems tailored to these vehicles.

To develop an optimal hydrogen fuel cell system for a specific type of construction equipment, it is essential to create an operating mode tailored to the target equipment. For commercial vehicles, such as those similar to mixer trucks, numerous studies have been conducted to analyze power characteristics and develop driving modes (Amirjamshidi et al., 2015; Zhang et al., 2021). However, mixer trucks possess unique features that differ significantly from those of standard commercial vehicles, including the operation of a rotating concrete drum while driving and performing concrete discharge tasks at construction sites. These distinctive characteristics create substantial limitations when applying the findings of previous research on commercial vehicles to mixer trucks. Some studies have analyzed the driving patterns of mixer trucks, but they are primarily confined to urban environments (Huang et al., 2021). Mixer trucks have the specific purpose of transporting concrete before it sets, which requires concrete plants to be situated in suburban areas, generally not far from urban centers. Consequently, a substantial portion of actual mixer truck driving occurs in suburban environments, and driving modes that reflect their driving characteristics naturally embody the traits of suburban driving.

However, few studies have considered these unique operational characteristics of mixer trucks. Developing driving modes that incorporate the unique operational features of mixer trucks can play a key role in facilitating and accelerating the design and research of future powertrains. Therefore, such research is essential and should be actively pursued.

Representative methodologies for developing driving modes include random-selection (Arun et al., 2017; Hung et al., 2007; Kamble et al., 2009; Seers et al., 2015; Tong et al., 2019), segment-based methods (Mayakuntla et al., 2018), clustering (Mohammadnazar et al., 2024; Berzi et al., 2016; Hull et al., 2024; Shen et al., 2018; Yanzhao et al., 2024; Tong et al., 2024), and Markov Chain Monte Carlo (MCMC) methods (Bishop et al., 2012; Bishop et al., 2024). Hung et al. (2007) utilized the micro-trip approach, defining operating data between idle and the next idle as a single trip, and developed a driving mode by randomly combining these trips. The model was then validated by evaluating parameters such as speed and acceleration (Tong et al., 2019). Mayakuntla et al. (2018) employed a segment-based method to define identification trips under specific conditions, selected the most representative parameters through component analysis, and subsequently developed a driving mode. Hull et al. (2024) analyzed operating data using the micro-trip method and developed a driving mode through clustering techniques. Bishop et al. (2012) adopted the MCMC method to develop a driving mode by utilizing data-based probability vectors without separating the speed and time data of the vehicle.

In this study, to incorporate as many real-world operating characteristics of mixer trucks as possible, operating data were collected during actual commercial driving without imposing any limiting conditions. Data collection was conducted over a three-month period for two mixer trucks.

Fig. 1 Mixer trucks for acquiring the experiment data

Table 1 Detailed specifications of 6-rube class mixer trucks

Items	Specification
Manufacturer	TATA DAEWOO
Model	New Prima 6 m^3 Mixer
Total weight (kg)	26,630
Engine displacement (cc)	8,700
Max power (HP)	400
Max torque (Nm)	1,700

Using the acquired operating data, various methodologies were evaluated for developing driving modes.

While clustering and MCMC-based methods have demonstrated robust performance for small to medium-sized datasets, their application to large-scale data remains constrained by the need for extensive parameter tuning and substantial computational resources. In contrast, the random-selection combined micro-trip approach offers a more streamlined and scalable alternative. By segmenting and reconstructing representative driving behaviors with minimal computational overhead, this method inherently enhances both efficiency and reproducibility. Such advantages become particularly evident when analyzing extensive real-world datasets, as in the present study, where computational tractability and data-handling scalability are critical to ensuring methodological robustness.

Furthermore, although the micro-trip method is commonly associated with urban driving conditions due to its ability to capture frequent stop-and-go behavior, it is also applicable to suburban driving patterns, particularly in the context of ready-mixed concrete mixer trucks. These trucks frequently decelerate and accelerate due to traffic signals, intersections, and loading/unloading tasks at construction sites located in decentralized areas. As such, their driving behavior naturally includes the segmented driving dynamics that the micro-trip method is designed to capture. Considering these factors, the micro-trip method combined with a random-selection technique was deemed the most suitable approach for this study. In addition, power characteristics were analyzed to evaluate the feasibility of designing and applying electric or hydrogen fuel cell powertrain systems in future applications.

2 Methodology

Diesel engine mixer trucks and hydrogen fuel cell electric mixer trucks utilize fundamentally different power sources. However, their power demand characteristics are nearly identical when operating the same type of equipment. In this study, driving data obtained from diesel engine mixer trucks were thoroughly analyzed with respect to power demand and driving behavior. The operating data acquired from diesel engine mixer trucks can be effectively used to evaluate whether a hydrogen fuel cell system can meet the required power characteristics.

The driving data were collected from the same model of 6-rube class mixer trucks, as shown in Fig. 1. These mixer trucks were powered by an FPT CURSOR 9 engine, with their detailed specifications provided in Table 1. The development process of the driving modes consisted of five steps:

1) Mixer truck data acquisition.
2) Analysis of mixer truck operating characteristics and selection of reference cycle.
3) Micro-trip method analysis and generation of candidate driving cycles.
4) Selection of representative driving mode.
5) Synchronization of power demand.

A detailed description of each step is provided below.

2.1 Mixer Truck Data Acquistion

Concrete mixer trucks are specialized commercial vehicles that transport ready-mixed concrete from batching plants to construction sites. Due to the nature of concrete, which begins to set within a limited time after mixing, these trucks

typically operate within a restricted delivery radius, usually within 15–25 km from the plant. As a result, they follow semi-fixed routes between the plant and various construction sites located in nearby suburban or mixed-use areas. Unlike public transport vehicles that operate on fixed schedules and fixed routes, mixer trucks are dispatched on a demand-driven basis, depending on construction project timelines. Their daily operation involves multiple round trips, but the departure times, destinations, and return paths can vary. However, because construction work is generally carried out during standard working hours, mixer truck activity is usually limited to daytime operation.

A conventional driving mode consists of a speed profile versus time (Kaymaz et al., 2019). However, to verify the applicability of fuel cell system development, demand power data is essential. Therefore, additional parameters such as the speed and location of the mixer truck, engine speed, and torque percentage were collected. Engine and speed data were measured via the OBD2 port using an IMC BUSDAQ-flex with the SAE J1939 protocol at a 10 Hz sampling frequency. The location of the mixer truck was recorded using a precision GPS device from Garmin, operating at a 5 Hz sampling frequency, and OBD data was changed to 5 Hz to synchronize. The mixer trucks were operated under typical conditions without any imposed restrictions (Seers et al., 2015). The driving data were collected over a three-month period, covering a total driving distance of 18,600 km and an operating time of 736 h. The results of the mixer trucks' locations, based on the acquired driving data, are shown in Fig. 2. In Fig. 2, the yellow line represents the driving path of the mixer trucks, the green point indicates the concrete plant and the red points represent the construction sites. Concrete plants exhibit a decentralized supply characteristic by region because ready-mixed concrete hardens within a specific period after transportation. As a result, construction sites are distributed within a certain distance around the concrete plant. the process of departing from the concrete plant, arriving at a specific construction site, and then returning to the concrete plant constitutes a single driving cycle. Concrete plants are mostly located near the outskirts of cities to supply concrete to the city or suburbs before it hardens. Data was collected from the Jeonbuk-do region in South Korea among concrete plants in suburban areas. When looking at the height of the topography, the Jeonbuk -do region is composed of a relatively large number of plains, with flat areas below 100 m accounting for 51% of the total, mountainous areas ranging from 100 to 500 m accounting for 33%, 500-1,000 m accounting for 13%, and highlands over 1,000 m accounting for 3%. (Jeonbuk-do website, 2025). Through the collected GPS data, the average altitude difference was calculated to be 0.08 m and the average slope angle was 0.6 °degrees. Based on this value, the road slope parameter was judged to be of little significance and was excluded.

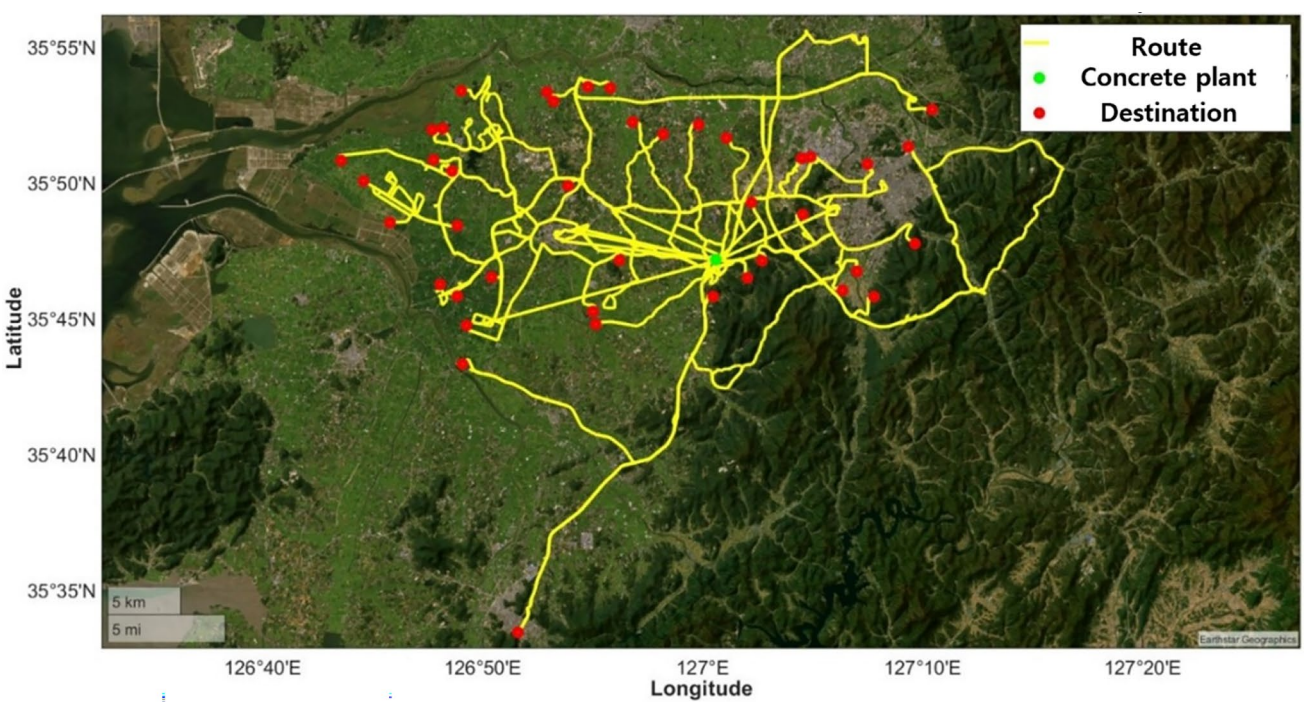


Fig. 2 Driving path of mixer trucks during data acquisition

2.2 Analysis of Mixer Truck Operating Characteristics and Selection of Reference Cycle

To conduct a detailed analysis of the acquired data, the driving data were classified into two categories: data corresponding to the movement from the concrete plant to the construction site, and data corresponding to the return from the construction site to the concrete plant. The data measured during the movement to the construction site were designated as the transport condition, and the data collected during the return to the concrete plant were referred to as the return condition. The speed and acceleration were analyzed for the transport and return conditions, as well as for the entire driving dataset. The speed distributions at 10 km/h intervals for the entire dataset and for each individual trip were compared, and the error for each bin was calculated. The mean absolute error (MAE) defined in Eq. 1 was then used to calculate the MAE for each individual trip. For both the transport and return conditions, the individual trip with the lowest MAE was selected as the reference cycle.

$$MAE = \frac{1}{n}\sum |\hat{y} - y| \tag{1}$$

2.3 Micro-trip Method Analysis and Generation of Candidate Driving Cycles

The obtained speed data was post-processed by classifying it into trips as shown in Fig. 3 using the micro-trip method. The micro-trip approach is a method that segments the entire driving dataset into short, continuous driving intervals known as micro-trips.

Each micro-trip is defined as a continuous driving segment that begins when the vehicle starts moving (speed > 0 km/h) and ends when it comes to a complete stop (speed = 0 km/h). The idling period before the micro-trip is not included. This definition is suitable for representing the active motion segments relevant to power demand analysis.

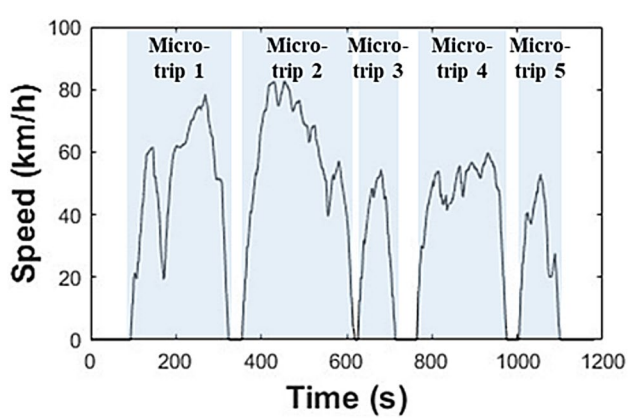


Fig. 3 Micro-trip method

The reference cycle was segmented into micro-trips, and for each micro-trip, the relative positive acceleration (RPA) and average speed (Vm) indicators used to closely examine acceleration and deceleration behaviors (Fernandes et al., 2024) were calculated. RPA is defined as shown in Eq. 2. The entire dataset was similarly segmented into micro-trips, and RPA and Vm were computed for each one. Based on these calculated values, the errors between the RPA and Vm of the reference cycle micro-trips and those of the entire dataset micro-trips were determined. For each micro-trip in the reference cycle, micro-trips from the entire dataset with an error of less than 5% were classified as candidate micro-trips. This selection criterion is consistent with the methodologies adopted in previous studies, such as Tong et al. (2019) and Arun et al. (2017), where similar error tolerances were applied for micro-trip matching. Candidate micro-trips were sampled using a random selection method, which is a commonly used technique in data sampling for driving mode analysis. This method helps reduce sampling bias and improves the generalizability of the results. These candidate micro-trips were then randomly combined to generate multiple candidate driving cycles.

$$RPA = \frac{\int_0^T \left(v_i \times a_i^+\right) \cdot dt}{x} \tag{2}$$

2.4 Selection of Representative Driving Mode

To select a representative driving mode among the generated candidate modes, it is essential to verify whether the mode exhibits characteristics similar to the average results of the entire driving dataset. In Europe, the real driving emissions (RDE) test defines the speed ranges for N3-class vehicles as below 45 km/h for urban areas, 45–70 km/h for suburban areas, and over 70 km/h for highway driving. In this study, based on the RDE test criteria for N3-class vehicles, the speed distribution was divided into three categories: Low (below 45 km/h), Mid (45–70 km/h), and High (above 70 km/h).

When selecting the representative driving mode, the speed distribution across each range and the average speed were used as comparison parameters. For each generated candidate driving mode, the speed distribution classified into three ranges and the average speed were calculated. Similarly, the speed distribution and overall average speed for the entire driving dataset were computed and compared. The errors in the speed distribution across the three ranges and the average speed were calculated, and the MAE defined in Eq. (1) was used to determine the MAE for each candidate driving mode. The candidate driving cycle with

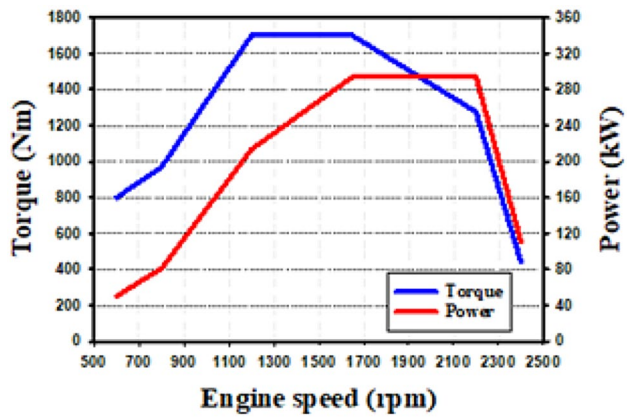


Fig. 4 Engine full-load performance curve

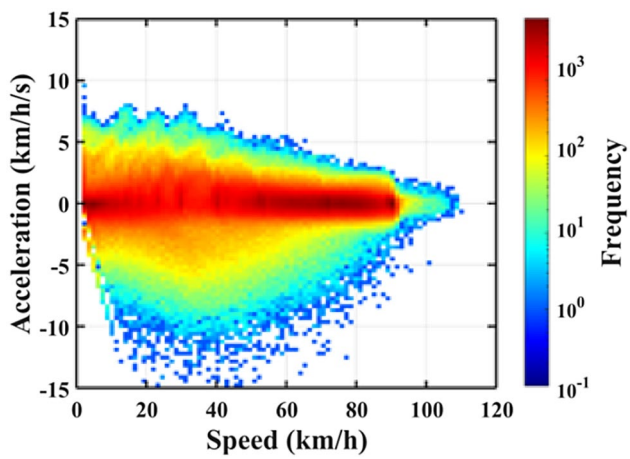


Fig. 5 Speed-based acceleration and frequency distributions of the entire data

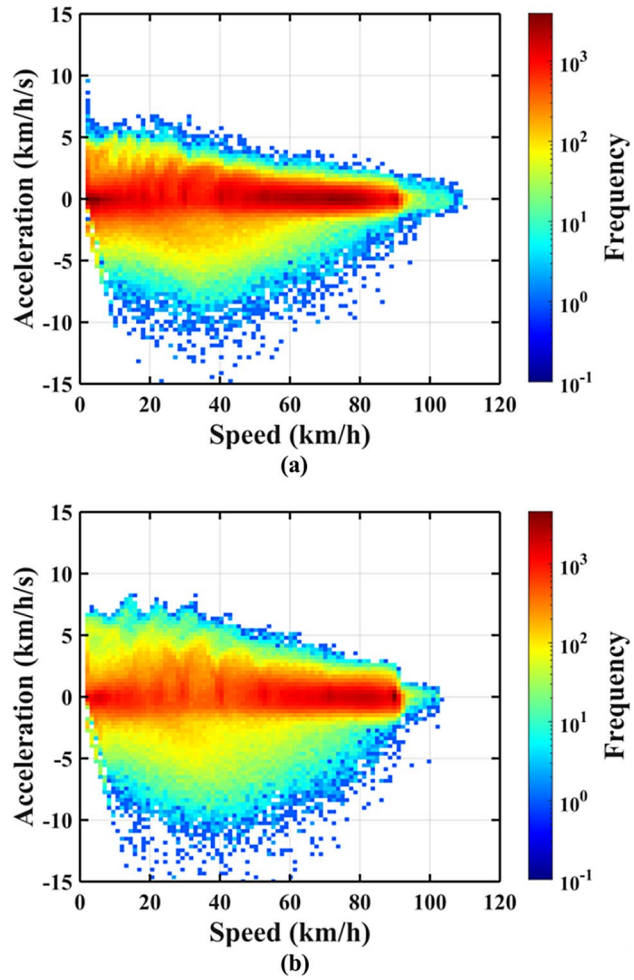


Fig. 6 Speed-based acceleration and frequency distributions **a** transport condition, **b** return condition

the lowest MAE was considered to most closely match the overall average results and was selected as the representative driving mode under both the transport and return conditions.

2.5 Synchronization of Power Demand Characteristics

The power demand characteristics are not only necessary for developing driving modes of mixer trucks, but also essential for verifying the future applicability of hydrogen fuel cell power systems. However, the representative operating modes for each condition are primarily based on speed profiles and time. To derive the power demand characteristics from the representative operating modes, the simultaneously measured engine speed and torque ratio data for each trip were compiled. The engine torque ratio data were converted into torque values in Nm using the engine full-load performance curve, as shown in Fig. 4 (FPT official web site, 2025). Based on the engine speed and torque, the power demand was calculated. Finally, the calculated power demand was synchronized with the speed profiles to derive the driving modes in terms of power demand.

3 Results

3.1 Driving Characteristics

Figure 5 presents the acceleration and frequency distributions as a function of vehicle speed based on the entire driving data of the mixer truck. Across the entire speed range of the mixer trucks, acceleration values of zero or near 0 accounted for the highest proportion. This indicates that steady driving constituted a significant portion of the overall driving. To provide a more detailed analysis, Fig. 6 presents the acceleration and frequency distributions classified into transport and return mode. During the return condition, the mixer trucks exhibited higher acceleration values more frequently than in the transport condition. Additionally, the

proportion of steady driving (0 or near 0 acceleration) was higher during transport than during return. The deceleration patterns were similar under both conditions. It is well-established that ready-mixed concrete inside the drum of a mixer truck hardens after a certain period. As a result, mixer trucks typically discharge all the loaded concrete at the construction site. Consequently, the mixer truck is lighter during the return condition compared to the transport condition. This reduction in weight results in brisker acceleration during return trips due to the increased engine power-to-weight ratio.

The 10 km/h interval speed distribution proportion result for transport and return modes of mixer trucks are presented in Fig. 7. A clear difference in speed distribution was observed between the transport and return conditions. mixer trucks tended to operate at higher speeds during return trips due to the increased engine power-to-weight ratio. Additionally, owing to the characteristics of suburban areas such as low traffic congestion and a high proportion of motorways the speed distribution exceeding 70 km/h was significantly higher during return trips compared to transport trips.

Power demand characteristics are essential for verifying the applicability of hydrogen fuel cell systems. As a preliminary step in power demand analysis, the torque distribution according to engine speed for the entire driving dataset is presented in Fig. 8. The engine frequently operated below 1,400 RPM in terms of engine speed and up to maximum torque in terms of engine torque. This driving is considered to result from the base power demand required to continuously rotate the concrete in the mixer truck, combined with the high proportion of low-speed driving.

The results of power demand are presented in Fig. 9. Under both transport and return conditions, more than 50% of the total operating occurred within segments where the power demand was 30 kW or lower. When Comparing transport and return, the proportion of power demand exceeding 120 kW was higher during transport than during return. The drum of the mixer truck is mechanically driven by the engine's Power Take-Off (PTO) system, and the engine torque measured during actual operation inherently includes the load associated with drum rotation. In the Transport mode, when concrete is loaded, the drum continuously rotates to prevent hardening, resulting in a relatively higher load. In contrast, during the Return mode, the absence of concrete eliminates the need for drum rotation, leading to a comparatively lower load on the engine. Since mixer trucks follow the same route for both transport and return condition, they maintain a consistent level of acceleration to align with traffic flow. However, in the transport condition, the increased overall weight due to the loaded ready-mixed concrete results in a higher power demand to achieve the same level of acceleration.

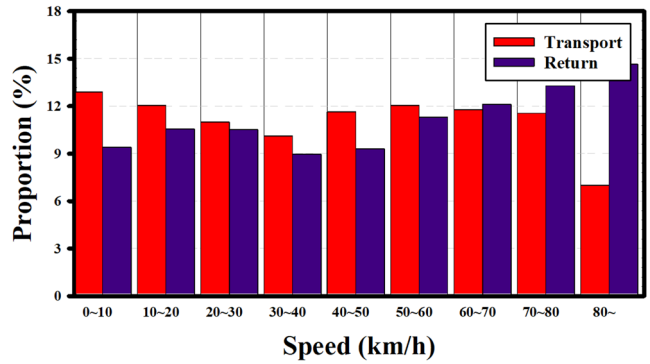


Fig. 7 Distribution of driving proportion by mixer truck speed on transport and return conditions

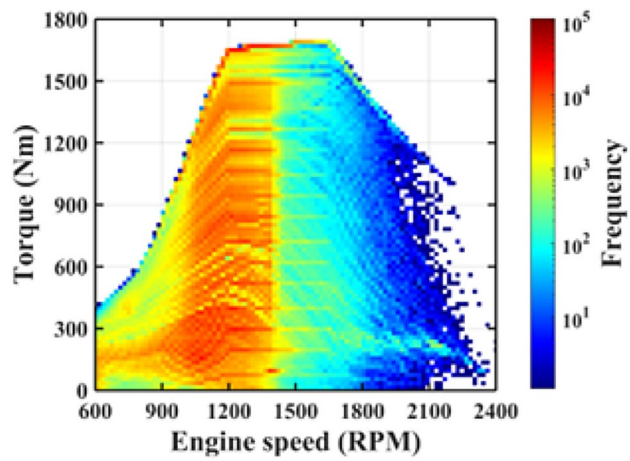


Fig. 8 Engine speed-based torque and frequency distributions

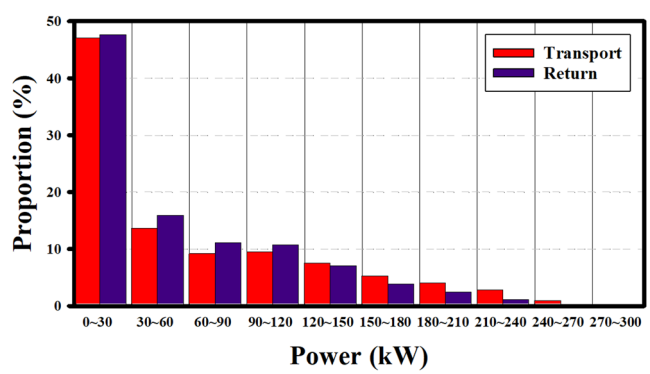


Fig. 9 Power demand distribution of mixer trucks under transport and return conditions

3.2 Generating Driving Modes

Figure 10 presents the process of extracting the reference cycle. A single driving cycle was defined as traveling from the concrete plant to the construction site for delivery and then returning from the construction site back to the plant. To select a representative cycle among multiple driving cycles, the speed distributions were compared. Specifically,

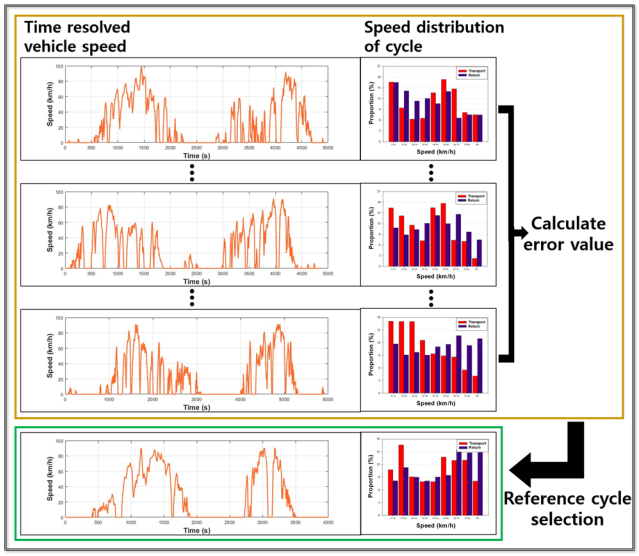
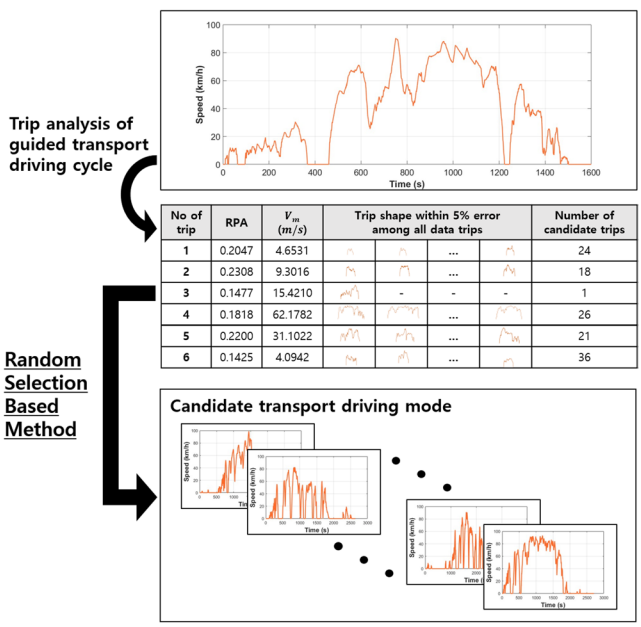


Fig. 10 Speed distribution for each travel to select the reference cycle

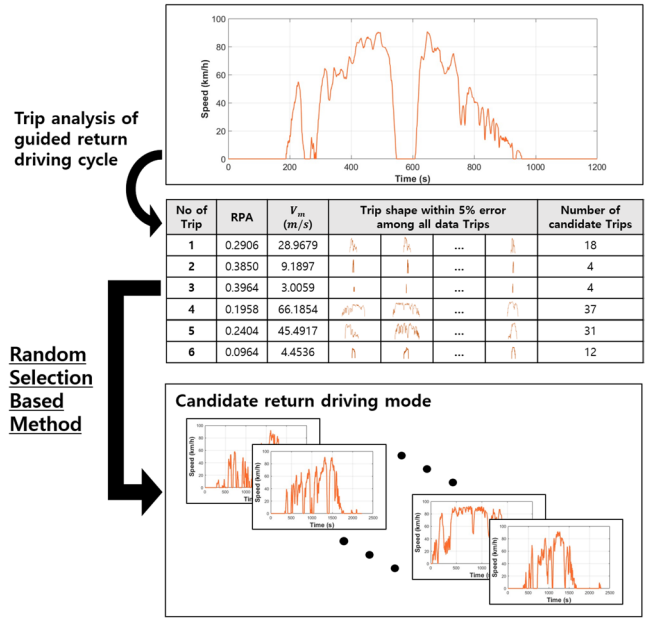


No of trip	RPA	V_m (m/s)	Trip shape within 5% error among all data trips				Number of candidate trips
1	0.2047	4.6531			...		24
2	0.2308	9.3016			...		18
3	0.1477	15.4210		-	-	-	1
4	0.1818	62.1782			...		26
5	0.2200	31.1022			...		21
6	0.1425	4.0942			...		36

Fig. 11 Process of selecting candidate modes of the mixer truck for transport condition

the mean absolute error (MAE) was calculated by comparing the speed distribution rates at 10 km/h intervals between the entire dataset and each individual driving cycle. The driving cycle with the lowest MAE value was identified as the most representative and was selected as the reference cycle.

Figures 11 and 12 present an analysis of the transportation and return mode reference cycles using the micro-trip method and depict the process of generating multiple candidate driving modes. The reference cycle for transport consisted of six micro-trips. RPA and Vm were calculated for



No of Trip	RPA	V_m (m/s)	Trip shape within 5% error among all data Trips				Number of candidate Trips
1	0.2906	28.9679			...		18
2	0.3850	9.1897			...		4
3	0.3964	3.0059			...		4
4	0.1958	66.1854			...		37
5	0.2404	45.4917			...		31
6	0.0964	4.4536			...		12

Fig. 12 Process of selecting candidate modes of the mixer truck for return condition

Table 2 Driving characteristics of the mixer truck during transport mode

Transport			
	Low	Mid	High
Proportion (%)	51.9	29.5	18.6
Average speed (km/h)	21.9	57.5	78.7
Time (min)	10.9	6.2	4.0
Travel time (min)	21.1		

each of the six micro-trips in the reference cycle. Based on the calculated values, six micro-trips were compared with individual micro-trips extracted from the entire data set, and micro-trips with an error of less than 5% were classified as candidate micro-trips of the reference cycle. Among the six classified candidate micro-trips, one was randomly selected from each group and recombined to generate candidate driving cycles. Through this process, a total of 8,491,392 candidate driving cycles were generated under transportation conditions. The same process was applied to the return conditions, resulting in a total of 3,964,032 candidate driving cycles.

3.3 Selecting the Representative Driving Mode

Tables 2 and 3 show the speed distribution results for the low, mid, and high-speed ranges based on the entire dataset under transportation and return conditions. The results indicated that the proportion of driving speeds exceeding 70 km/h(High) increased significantly during return trips. Conversely, the proportion of driving speeds below 45 km/h(Low) decreased. It is important to note that mixer

Table 3 Driving characteristics of the mixer truck during return mode

Return			
	Low	Mid	High
Proportion (%)	43.9	28.1	28.0
Average speed (km/h)	22.4	58.1	80.6
Time (min)	7.8	5.1	5.1
Travel time (min)	18.0		

Table 4 Idle proportion and duration of the mixer truck on a travel for each condition

	Transport	Return
Proportion (%)	19.7	17.6
Duration (min)	5.19	3.85

trucks typically used the same path for transport and return, meaning the road conditions were consistent for both directions. This change in the speed distribution was attributed to differences in available acceleration resulting from variations in the power-to-weight ratio of the mixer truck. The average travel time was 21 min and 7 s for transport and 17 min and 59 s for return, as the average speed during return trips was higher compared to transport trips. The speed distribution, average speed, and travel time by section comprising Low, Mid, and High segments during both transport and return were used as parameters to verify the development of the driving modes.

The speed distribution across low, mid, and high-speed ranges was also analyzed for each generated candidate driving cycle. In addition, the MAE between the low, medium, and high-speed ratios, as well as the average speed, of each candidate driving cycle and those of the entire dataset was compared. The candidate driving cycle with the smallest MAE was selected as the representative driving mode for each condition. For the transportation condition, a representative transportation mode with a MAE of 1.96% was selected, while for the return condition, a representative return mode with a MAE of 0.83% was chosen. During the random selection process, idle operation was excluded. Subsequently, the idle operation statistics obtained from the full dataset were analyzed, as shown in Table 4, and were integrated between trips in the selected representative modes. The average power was 6.28 kW under transportation conditions and 6.11 kW under return conditions, indicating that the transportation mode exhibited slightly higher power.

The representative driving modes constructed based on the mixer truck speed profiles over time are presented in the lower part of Fig. 13 for both the transport and return modes. To utilize these operating modes not only for analyzing the operational characteristics of conventional mixer trucks but also for the development or design of hydrogen fuel cell systems, it is necessary to synchronize the power demand profile with the speed profile. The power demand profiles synchronized with the mixer truck speeds were derived using the engine RPM and torque data that constitute the representative modes. The resulting power demand profiles are presented in upper part of Fig. 13.

A comparison of the developed mixer truck driving mode with representative standard driving cycles (e.g., NRTC, WLTC, FTP, NEDC, and JC08) is presented on Fig. 14. The comparison was based on RPA and average driving speed. As shown in the figure, the mixer truck modes ("mixer transport" and "mixer return") are located in the high RPA and low speed ranges, indicating frequent acceleration and deceleration behavior under real driving conditions.

Power demand was analyzed to determine the fuel cell stack power and battery capacity necessary for developing a hydrogen fuel cell system. A moving average analysis methodology was applied to the representative driving modes of power demand. The average and maximum power demand based on the size of the moving average window is shown in Fig. 15 for both transport and return conditions. the mixer truck required a peak power demand of 282 kW under both transport and return conditions. Consequently, the fuel cell power system for the 6-rube class mixer truck

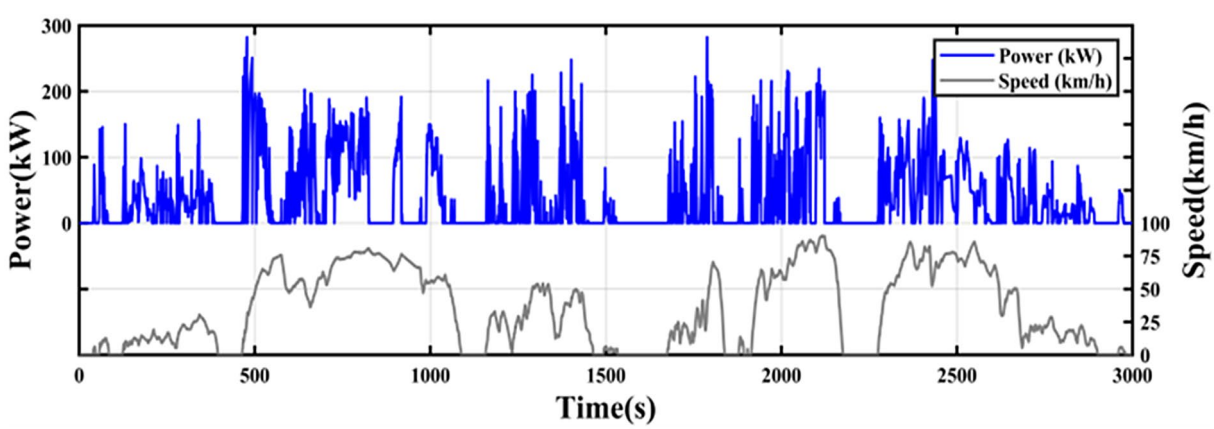


Fig. 13 The representative driving mode

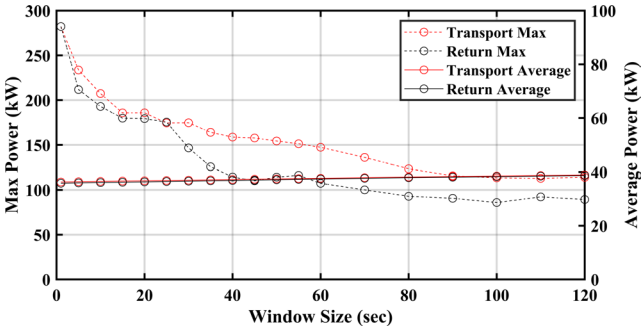


Fig. 14 RPA against average driving speed correlation between the developed mixer-truck modes and conventional standard driving cycles

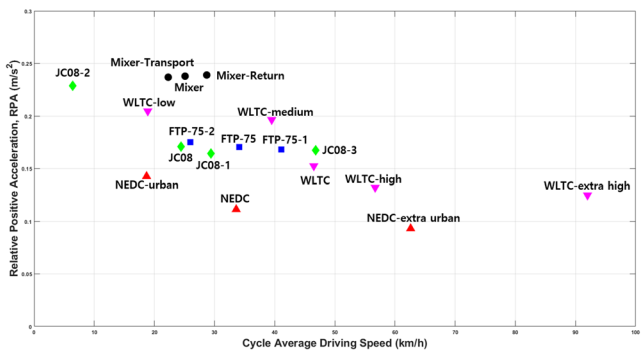


Fig. 15 Maximum and average power against moving window size

must be capable of providing a peak instantaneous power of 282 kW. Furthermore, the feasibility of increasing battery capacity to reduce the required power of the fuel cell stack was demonstrated by the observed reduction in maximum power within the moving average window as the window size increased, as shown in Fig. 15. The average and maximum power for transport and return conditions, based on real-time data and a 30-second moving average window, are presented in Table 5. The average power and maximum power for transport and return conditions, based on real-time data and a 30-second moving average window, were analyzed. These metrics provide insight into the power demand profiles under varying operational conditions, highlighting the distinctions between instantaneous and smoothed power requirements.

4 Conclusion

In this study, actual driving data were acquired from mixer trucks operating in suburban areas, and driving modes were developed using the random-selection method. These developed driving modes were further derived as power demand profiles to enable verification of their applicability in the development of hydrogen fuel cell systems. This research contributes to determining the optimal power of the hydrogen fuel cell stack and battery capacity for integrating hydrogen fuel cell systems into mixer trucks. Future work will focus on developing driving modes for concrete pump trucks, which are frequently used alongside mixer trucks at construction sites. The conclusions of this study are summarized as follows:

1) Analysis of acceleration, speed, and power demand revealed that mixer trucks exhibited more frequent and stronger acceleration, higher speeds, and lower power demand during return condition compare to transport condition due to reduced weight.
2) A representative reference cycle was selected via MAE-based speed distribution comparison, and 8,491,392 and 3,964,032 candidate driving cycles were generated for transport and return conditions, respectively, by recombining micro-trips.
3) Representative driving modes for transport and return were selected based on speed distribution and MAE analysis, and synchronized power demand profiles were constructed using engine data to enable application in hydrogen fuel cell system development.
4) Power demand analysis revealed a peak power requirement of 282 kW for the mixer truck, and demonstrated the potential for increasing battery capacity to reduce fuel cell power needs through moving average analysis.

This study developed driving modes based on data collected and analyzed from 6-rube class mixer trucks. Therefore, future studies focusing on mixer trucks of different capacities are expected to provide a broader understanding of their driving characteristics and enable more diverse mode development.

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Table 5 The average power and maximum power for transport and return conditions, based on real-time data and a 30 s moving average window

	Real-time during the driving		Moving average in 30 s	
	Average power (kW)	Max power (kW)	Average power (kW)	Max power (kW)
Transport	36.3	282.2	36.9	174.7
Return	35.8	282.2	36.6	146.7

Declarations

Competing interests The authors declare that they have no competing interests.

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